SEAT No.________ No. of Printed Pages : 2

SARDAR PATEL UNIVERSITY, VALLABH VIDYANAGAR

[53] **BCA (2^{nd} Semester) External Examination (Batch-2019)**

16-08-2019

Subject: US02SBCA21 (Mathematics)

Time: 2:00 to 4:00 p.m. **Total Marks: 35**

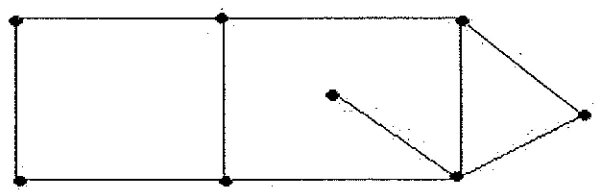
Q.1 Multiple Choice Questions **[05]**

1. A graph with one vertex and no edges is called.........
   (A) trivial graph (B) Multigraph
   (C) Null graph (D) None
2. A graph with no vertices and no edges is called__________
   (A) trivial graph (B) Multigraph
   (C) Null graph (D) Multiple edges
3. A map is a particular ____________ representation of a finite planar multigraph.
   (A) Planar (B) Non-planar
   (C) Complete (D) simple
4. In a coloring of a map M, one required to color the ________
   (A) Edges (B) Vertices
   (C) regions (D) None
5. Which of the following is not the measure of Dispersion
   (A) Range (B) Mode
   (C) Both (A) & (B) (D) None

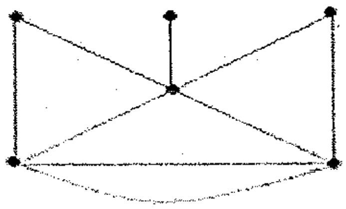
Q.2 Short Questions (Attempt any Five) **[10]**

1. Draw a picture of each of the following graphs, $G_1 = (V_1,E_1)$, where $V_1 = \{a, b, c, d, e\}$ and $E_1 = \{ab, bc, ac, ad, de\}$.
2. Define Complete Graph with example.
3. Define trivial graph and Multigraph
4. The intelligence quotients (IQ's) of 10 boys is given below:
   70, 120, 110, 101, 88, 83, 95, 98, 107, 100 Find the Mean IQ.
5. List out advantages of Mean and Mode.
6. List out types of Matrix with example of each.
7. If $u = (1,2,3,-4)$, $v = (5,-6,7,8)$, $w= (1,-2,5,3)$ and $k=3$
   then find : (i) $(u + v) \bullet w$ (ii) $u \bullet v + v \bullet w$
   (iii) $k\,(u \bullet v)$ (iv) $(ku) \bullet v$
8. Determine k such that $\| u \| = \sqrt{39}$ where $u = (1,k,-3,-5,1)$

Q.3 (A) Find out possible cycle and closed path from given graph and degree of vertex of each node. **[05]**

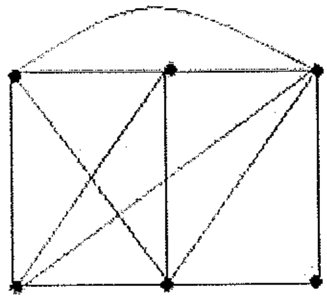


PTO

(B)  **[05]**

Find the degree of Regions from given Graph as well as find degree of vertex.

OR

Q.3 (A)  **[05]**

Find out chromatic number from given graph.

(B) Consider u = (1,-2,3,4), v = (-2,4,5,-3) w= (5,-5,-3,4) and z = (2,7,4,-2). **[05]**
Find the followings:

(i) $u + v$ (ii) $-3u$ (ix) $5w - 9u + 3v$
(iii) $-w$ (iv) $3u - 4w$ (x) $9w + 4v - 1u$
(v) $2u - 4w + 3v - 2z$ (vi) $2u + 5z$
(vii) $u \bullet v$ (viii) $v \bullet w$

Q.4 Find the value of mean, median and mode from the given data. **[06]**

(A)

Wt. in kg	93-97	98-102	103-107	108-112	113-117	118-122	123-127	128-132
No. of Stud.	3	5	12	17	14	6	3	1

(B) **[04]**

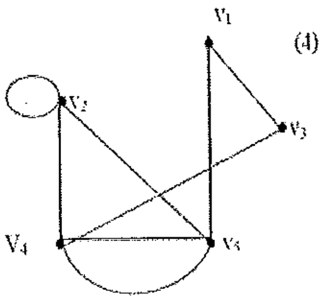
Find the inverse of A= $\begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{pmatrix}$

OR

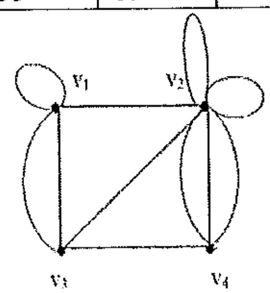
Q.4 (A) Find x, y, z and t, if $3\begin{pmatrix} x & y \\ z & t \end{pmatrix} = \begin{pmatrix} x & 6 \\ -1 & 2t \end{pmatrix} + \begin{pmatrix} 4 & x+y \\ z+t & 3 \end{pmatrix}$ **[04]**

(B) Draw histogram for following frequency distribution **[03]**

Variable	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Freq.	12	30	35	65	45	25	18

(C) 

(4) and (6)



Find Adjacent Matrix and Incidence Matrix from Given Graphs. **[03]**

—x—